

IPL Crunch '26 — Data Analytics Report

IPL Crunch '26 — Data Analytics Challenge A

Data-Driven Analysis of IPL Match Dynamics

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Objective

This project analyzes IPL ball-by-ball datasets to uncover patterns related to toss impact on match outcomes, match-winning phases, top-performing batters and bowlers, team consistency across seasons, and hidden insights revealed through data.

Tools Used

- Python
- Pandas
- Matplotlib
- Jupyter Notebook

Dataset Source

- Cricsheet IPL JSON datasets

Dataset Overview

Multiple IPL seasons were analyzed using ball-by-ball IPL datasets. The data was processed to generate match-level and player-level insights. Match phases were classified into:

- Powerplay (Overs 1–6)
- Middle Overs (7–15)
- Death Overs (16–20)

The goal was to transform raw IPL data into meaningful cricket insights.

Does Winning the Toss Matter?

Teams winning the toss also won the match in approximately 51.62% of completed games.

Key Insight:

The toss provides only a marginal advantage. Match execution and performance have a far greater impact on victory than pre-match conditions.

Which Match Phase Impacts Victory Most?

Powerplay Overs:

Important for early momentum but less influential overall.

Middle Overs:

Strongly influence game control and momentum building.

Death Overs:

Show the strongest relationship with winning matches. Finishing strength and late-game control significantly affect outcomes.

Key Insight:

Death overs have the highest impact on IPL match victories.

Top IPL Batters

Virat Kohli emerged as the leading run scorer in the dataset, demonstrating exceptional consistency and longevity across seasons.

Key Insight:

Long-term consistency separates elite IPL batters from short-term peak performers.

Top IPL Bowlers

Yuzvendra Chahal emerged as the leading wicket taker in the analyzed dataset.

Key Insight:

Successful IPL bowlers combine control, consistency, and wicket-taking ability.

Team Performance Trends

Chennai Super Kings and Mumbai Indians consistently remained among the strongest teams across seasons, while some newer franchises showed strong short-term spikes.

Key Insight:

Stable team structures and long-term consistency contribute heavily to sustained IPL success.

Most Surprising Insight

The most surprising finding was how little the toss explained match outcomes when compared with batting and bowling performance during the death overs.

This insight challenges common fan assumptions and highlights the importance of finishing strength in T20 cricket.

Conclusion

- Toss advantage exists but is relatively small.
- Death overs are the strongest predictor of victory.
- Elite IPL players succeed through consistency over multiple seasons.
- Data analytics helps uncover patterns beyond intuition.

This project demonstrates how sports analytics can transform raw cricket data into meaningful strategic insights.